

Python Automation

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What is Automation in Python?

- ▶ **Automation in Python** refers to using Python scripts to perform tasks automatically without manual intervention. These tasks can range from simple repetitive operations to complex workflows. Python is widely used for automation because of its:
 - ▶ **Simplicity and readability**
 - ▶ **Large ecosystem of libraries and tools**
 - ▶ **Cross-platform compatibility**

Examples of Automation in Python:

1.File and Folder Automation:

Renaming, moving, or deleting files.

Organizing directories automatically.

2.Web Scrapping:

Extracting data from websites using tools like BeautifulSoup or Selenium.

3.Data Processing:

Automating data analysis with pandas or NumPy.

4.Web Automation:

Filling forms, clicking buttons, and scraping content using Selenium.

5.Email and Messaging:

Sending emails or WhatsApp messages using smtplib, pywhatkit etc

6.Task Scheduling:

Running periodic scripts using schedule, cron, or Windows Task Scheduler.

7.APIs and Bots:

Creating bots for Telegram, Discord, or automating tasks via REST APIs

shutil Library Example

- ▶ Copy, move, delete files and folders
- ▶ `shutil.copy()`, `shutil.move()`, `shutil.rmtree()`
- ▶ Useful for file organization and cleanup

Other Useful Libraries for Automation

- ▶ os - File and directory handling
- ▶ requests, BeautifulSoup4 - Web automation and scraping
- ▶ openpyxl, pandas - Excel and CSV automation
- ▶ smtplib, email - Sending emails
- ▶ pyautogui - GUI automation
- ▶ pywhatkit - WhatsApp automation
- ▶ schedule - Task scheduling